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Web Technology

Quality Laboratory Manual

Experiment No. 02

Program Based on Typography and Background Properties of CSS3, and Animation Effect Using Transition



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Experiment No. 02

Title of Experiment: Program Based on Typography and Background Properties of CSS3, and Animation Effect Using Transition

Aim of Experiment: To design a web page using CSS3 typography and background properties and to implement animation effects using CSS transition.

System Requirements –

Operating System: Windows 10 or above

RAM: Minimum 4 GB

Software/s Needed for Experiment – (Chrome/Edge/Firefox), Text Editor (VS Code / Notepad++)

Experiment Objectives:

- To understand typography properties in CSS3.
- To apply background properties in web design.
- To implement transition effects for animation.
- To design an attractive and interactive web page using CSS3.

Experiment Outcomes:

- Apply CSS3 typography properties such as font, color, spacing, and alignment.
- Use CSS3 background properties including images, gradients, and positioning.
- Create smooth animation effects using CSS transitions.
- . Design visually appealing and interactive web pages using CSS3.

Theory:

CSS Transitions

CSS transitions allows you to change property values smoothly, over a given duration.

Mouse over the element below to see a CSS transition effect:

In this chapter you will learn about the following properties:

- transition
- transition-delay
- transition-duration
- transition-property
- transition-timing-function

How to Use CSS Transitions?

To create a transition effect, you must specify two things:

- the CSS property you want to add an effect to
- the duration of the effect

Note: If the duration part is not specified, the transition will have no effect, because the default value is 0.

The following example shows a 100px * 100px red <div> element. The <div> element has also specified a transition effect for the width property, with a duration of 2 seconds:

Example

```
div {  
  width: 100px;  
  height: 100px;  
  background: red;  
  transition: width 2s;  
}
```

The transition effect will start when the specified CSS property (width) changes value.

Now, let us specify a new value for the width property when a user mouses over the <div> element:

```
<!DOCTYPE html>  
<html>  
<head>  
<style>  
div {  
  width: 100px;  
  height: 100px;  
  background: red;  
  transition: width 2s;  
}  
div:hover {  
  width: 300px;  
}  
</style>  
</head>  
<body>  
<h1>The transition Property</h1>  
<p>Hover over the div element below, to see the transition effect:</p>  
<div></div>  
</body>  
</html>
```

Notice that when the cursor mouses out of the element, it will gradually change back to its original style.

Change Several Property Values

The following example adds a transition effect for both the width and height property, with a duration of 2 seconds for the width and 4 seconds for the height:

Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s, height 4s;
}
div:hover {
  width: 300px;
  height: 300px;
}
</style>
</head>
<body>
<h1>The transition Property</h1>
<p>Hover over the div element below, to see the transition effect:</p>
<div></div></body></html>
```

Specify the Speed Curve of the Transition

The transition-timing-function property specifies the speed curve of the transition effect.

The transition-timing-function property can have the following values:

- ease - specifies a transition effect with a slow start, then fast, then end slowly (this is default)
- linear - specifies a transition effect with the same speed from start to end
- ease-in - specifies a transition effect with a slow start
- ease-out - specifies a transition effect with a slow end
- ease-in-out - specifies a transition effect with a slow start and end
- cubic-bezier(n,n,n,n) - lets you define your own values in a cubic-bezier function

The following example shows some of the different speed curves that can be used:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s;
}
#div1 {transition-timing-function: linear;}
#div2 {transition-timing-function: ease;}
#div3 {transition-timing-function: ease-in;}
#div4 {transition-timing-function: ease-out;}
```

```
#div5 {transition-timing-function: ease-in-out;}
div:hover {
  width: 300px;
}
</style>
</head>
<body>
<h1>The transition-timing-function Property</h1>
<p>Hover over the div elements below, to see the different speed curves:</p>
<div id="div1">linear</div><br>
<div id="div2">ease</div><br>
<div id="div3">ease-in</div><br>
<div id="div4">ease-out</div><br>
<div id="div5">ease-in-out</div><br>
</body>
</html>
```

Delay the Transition Effect

The transition-delay property specifies a delay (in seconds) for the transition effect.

The following example has a 1 second delay before starting:

Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 3s;
  transition-delay: 1s;
}
div:hover {
  width: 300px;
}
</style>
</head>
<body>
<h1>The transition-delay Property</h1>
<p>Hover over the div element below, to see the transition effect:</p>
<div></div>
<p><b>Note:</b> The transition effect has a 1 second delay before starting.</p>
</body>
</html>
```

Transition + Transformation

The following example adds a transition effect to the transformation:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  transition: width 2s, height 2s, transform 2s;
}
div:hover {
  width: 300px;
  height: 300px;
  transform: rotate(180deg);
}
</style>
</head>
<body>
<h1>Transition + Transform</h1>
<p>Hover over the div element below:</p>
<div></div>
</body>
</html>
```

More Transition Examples

The CSS transition properties can be specified one by one, like this:

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  width: 100px;
  height: 100px;
  background: red;
  transition-property: width;
  transition-duration: 2s;
  transition-timing-function: linear;
  transition-delay: 1s;
}
div:hover {
  width: 300px;
}
</style>
</head>
<body>
<h1>The transition Properties Specified One by One</h1>
<p>Hover over the div element below, to see the transition effect:</p>
<div></div>
<p><b>Note:</b> The transition effect has a 1 second delay before starting.</p>
</body>
```

</html>

Program Code:

```
<!DOCTYPE html>
<html>
<head>
<title>CSS3 Typography, Background & Transition Demo</title>
<style>
body {
    font-family: Arial, sans-serif;
    background-color: #f5f5f5;
    margin: 0;
    padding: 0;
}
header {
    background: linear-gradient(45deg, #007BFF, #00C6FF);
    color: white;
    text-align: center;
    padding: 20px;
    font-size: 28px;
}
.content {
    padding: 20px;
    font-size: 18px;
    color: #333;
}
.box {
    width: 300px;
    height: 150px;
    background-image: url('https://via.placeholder.com/300');
    background-size: cover;
    background-position: center;
    border-radius: 10px;
    margin: 20px auto;
    text-align: center;
    color: white;
    padding-top: 60px;
}
.button {
    display: inline-block;
    padding: 12px 25px;
    background-color: #2a9d8f;
    color: white;
    text-decoration: none;
    border-radius: 5px;
    transition: all 0.5s ease;
}
.button:hover {
    background-color: #e76f51;
    transform: scale(1.1);
}
```

```
</style>
</head>
<body>
<header>CSS3 Typography, Background & Transition Effects</header>
<div class="content">
<p>This web page demonstrates CSS3 typography, background properties, and transition effects.</p>
</div>
<div class="box">Background Image Box</div>
<center><a href="#" class="button">Hover Me</a></center>
</body>
</html>
```

Task for Students: Designing a Modern Website for “Smart Learning Academy”

Task Requirements:

- 1) Typography (Text Styling)
- 2) Background Design
- 3) Transition Effects (Animation)
- 4) Web Page Layout

Submission Requirement:

- 1) HTML file (.html).
- 2) CSS file (.css) (optional).
- 3) Screenshot of output.

Observations:

1. Typography improved text readability.
2. Background properties enhanced visual design.
3. Transition effects created smooth animations.
4. CSS3 features made the web page interactive.

Conclusion:

This experiment demonstrated the use of CSS3 typography, background properties, and transition effects. These features help in designing modern, interactive, and visually appealing web pages.

Expected Oral Questions:

1. What is CSS typography?
2. Explain background properties in CSS3.
3. What is CSS transition?
4. Difference between transition and animation?

FAQs in Interview:

1. What is CSS3?
2. What is typography in CSS?
3. List some CSS typography properties.
4. What are background properties in CSS3?
5. Name different background properties in CSS.
6. What is CSS transition?
7. Write the syntax of CSS transition.
8. What is the difference between transition and animation in CSS?
9. What is the use of hover in CSS?
10. What are gradients in CSS3?
11. Why are transitions important in web design?
12. Give examples of CSS transition effects.
13. What is the difference between background-size: cover and contain?
14. What is responsive design?
15. Which CSS3 features did you use in your experiment?